BDA601 Assessment 2 Report

Visualisation and Model Development: Telco Customer Churn

Subject: BDA601 Big Data and Analytics
Assessment: Individual - Source Code and Report (1,000 words +/- 10%)

1. Introduction

Customer churn is the movement of subscribers from one service provider to another. Retaining customers is usually cheaper than acquiring new ones, and long-term customers tend to be less costly to serve and less sensitive to competitor offers. This report supports a decision-tree churn model built from the Kaggle Telco Customer Churn IBM sample (https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data). The full extract contains 7,043 customers and 21 attributes. Per Assessment 2 Task 1, five attributes were then removed (MonthlyCharges, OnlineSecurity, StreamingTV, InternetService and Partner), leaving 16 attributes with Churn as the target. The notebook covers data extraction, exploratory analysis, cleaning, feature selection and model development. This report focuses on (a) a missing-value strategy for the most important attribute and (b) interpretation of the churn analysis outcomes.

2. Handling Missing Values

The decision-tree model ranked Contract as the most important predictor by Gini importance (approximately 0.65), far ahead of tenure (about 0.10) and TechSupport (about 0.09). This is consistent with exploratory findings: month-to-month customers churn at about 42.7%, compared with 11.3% for one-year and 2.8% for two-year contracts. In the given dataset, Contract has no missing values. However, in real operational data a critical attribute such as Contract can be incomplete due to CRM gaps, migration errors or incomplete sign-up records.

If a substantial share of Contract values were missing, deleting those rows would be undesirable because Contract is the dominant split feature and row deletion would shrink the training base and bias retention analysis. A practical replacement method is mode imputation within meaningful customer segments: for each incomplete record, impute Contract using the most frequent contract type among customers with similar tenure band and payment method (fallback to the global mode if a segment is too small). Contract is categorical, so the mode preserves valid category labels and avoids inventing ordinal scores. Where richer auxiliary data exist, a secondary classifier trained on complete cases could predict Contract before the main churn model; for this assessment, segmented mode imputation is transparent, reproducible and aligned with decision-tree workflows. The notebook demonstrates the impact of this idea by simulating 30% missingness in Contract and restoring values with the training-set mode, then re-fitting the tree to confirm the pipeline remains usable.

3. Interpretation of Churn Analysis

3.1 Effectiveness of the churn analysis

On the held-out test set (20% stratified sample), the tuned decision tree (max_depth = 6, min_samples_leaf = 20, balanced class weights) achieved approximately 73.2% overall accuracy and a ROC-AUC of about 0.81. In other words, the model assigned the correct churn label for roughly three-quarters of customers. Because only about 26.5% of customers churn, a trivial classifier that always predicts No would already reach about 73.5% accuracy; therefore raw accuracy alone is only borderline relative to that baseline.

Effectiveness is more meaningful when judged by detection of actual churners. Churn recall was approximately 76% (precision about 50%), meaning the model correctly flagged about three in four customers who left, at the cost of some false alarms. For retention planning this trade-off is often acceptable: contacting a customer who would have stayed is usually cheaper than missing a high-risk customer. Overall, the analysis is a satisfactory operational starting point for targeting, provided campaigns are designed around probability scores and cost-sensitive thresholds rather than accuracy alone. It is not yet a fully optimised production model, but it is interpretable and better than chance for ranking risk (AUC about 0.81).

3.2 Who is churning and what drives churn

Churners are disproportionately short-tenure, flexible-contract customers. Mean tenure for churners is about 18 months versus about 38 months for retained customers. Month-to-month contracts dominate risk (42.7% churn). Payment behaviour also matters: electronic-check users churn at about 45.3%, well above automatic bank transfer (about 16.7%) and credit card (about 15.2%). Customers without TechSupport churn at about 41.6% versus about 15.2% with support. Paperless billing (about 33.6% versus 16.3%) and absence of dependents (about 31.3% versus 15.5%) further elevate risk. Collectively, the profile is a relatively new, month-to-month subscriber with limited support add-ons and a higher-friction payment method. The tree's root importance on Contract indicates that contractual lock-in is the primary structural driver; tenure and support features refine risk inside contract groups. Business drivers therefore combine switching ease (no long-term commitment), weaker product stickiness (no tech support) and billing friction - not merely demographics such as gender, which showed little importance.

3.3 Improving accuracy of the churn analysis

Model development choices materially shaped outcomes. Cleaning blank TotalCharges values (tenure = 0 customers) by setting charges to zero prevented type errors and kept new customers in the sample. Removing customerID avoided leakage of an identifier. Stratified train/validation/test splits preserved class balance, and depth tuning on validation reduced overfitting relative to an unrestricted tree. Balanced class weights lowered pure accuracy slightly versus an unweighted tree but improved sensitivity to the minority churn class - appropriate for retention use.

Handling missing values in Contract via informed imputation would protect the model's strongest signal; poor imputation (for example random fills) would dilute the root splits and harm both accuracy and interpretability. Accuracy and ranking quality could be improved by: (1) restoring informative removed fields where policy allows (for example MonthlyCharges, InternetService) or engineering proxies; (2) using richer encodings (one-hot or target encoding) and ensemble methods (random forest / gradient boosting) while retaining a tree for explanation; (3) calibrating thresholds with business costs instead of the default 0.5 cut-off; (4) addressing class imbalance with resampling; and (5) adding behavioural time-series features (usage drops, complaint tickets). Together, these steps should lift discrimination beyond the current AUC of about 0.81 while preserving the clear who is churning narrative required for stakeholder communication.

4. Conclusion

A transparent decision-tree pipeline on the modified Telco dataset shows that contract type is the dominant churn signal, with tenure and support/billing attributes as secondary drivers. Test-set accuracy is about 73%, with stronger value in about 76% recall of churners and AUC near 0.81. A segmented mode-imputation strategy is recommended if Contract were heavily missing. With threshold tuning and richer features or ensembles, predictive performance can be improved without losing the interpretability that makes the analysis actionable for retention teams.

References

Kaggle.com. (2020). Telco customer churn - IBM sample data sets. https://www.kaggle.com/blastchar/telco-customer-churn